

Saddle-Dominated Capacity of the LSE Dense Associative Memory

Abstract

[Placeholder.] The log-sum-exp Dense Associative Memory [2] stores M unit vectors ξ^μ on the N -sphere. Ramsauer et al. proved $\alpha_c = 1/2$ at $T = 0$ by modelling the inter-pattern overlaps as Gaussian. Replacing the Gaussian by the exact spherical density opens an interior saddle in the partition sum at the golden ratio $\varphi^* = (\sqrt{5}-1)/2$ with value $g_{\max} = \frac{1}{2} \ln \varphi^* + \varphi^* \approx 0.3774$. The boundary becomes $\alpha_c(T) = 1 - g_{\max} - f_{\text{ret}}(T)$, so $\alpha_c(0) \approx 0.6226$, not 0.5 — a factor 1.25 in capacity. We confirm the bound with three Monte Carlo variants (direct, semismart, hybrid) at N up to 100.

1 Introduction

A Dense Associative Memory [1] stores M patterns $\xi^\mu \in \mathbb{R}^N$, $\mu = 1, \dots, M$, as minima of an energy function $H(\mathbf{x})$. Retrieval of ξ^1 from a noisy cue is the Metropolis dynamics of $\mathbf{x}$ under H at external temperature T . Both the patterns and the chain live on the N -sphere of radius $\sqrt{N}$.

The log-sum-exp (LSE) variant of Ramsauer et al. uses

$$H(\mathbf{x}) = -\beta_{\text{net}}^{-1} \ln \sum_{\mu=1}^M \exp(\beta_{\text{net}} \mathbf{x} \cdot \xi^\mu), \quad \beta_{\text{net}} = 1 \text{ throughout}, \quad (1)$$

and stores $M = e^{\alpha N}$ patterns at load $\alpha = \ln M/N$. The question is: what is the largest α at which the chain initialised at ξ^1 stays close to ξ^1 ?

Ramsauer et al. answered $\alpha_c = 1/2$ at $T = 0$. The proof modelled the inter-pattern overlap $\varphi_{1\mu} = \xi^1 \cdot \xi^\mu / N$ as a Gaussian with variance $1/N$. The Gaussian is the leading term of the exact spherical density

$$f(\varphi) \propto (1 - \varphi^2)^{(N-3)/2}, \quad \varphi \in [-1, 1], \quad (2)$$

near $\varphi = 0$, but the two densities differ near $\varphi = 1$. That tail matters: it sets the typical maximum of M samples and the saddle of the partition sum.

This paper does three things. First, we re-derive the boundary using the exact density (2). An interior saddle of the integrand appears at the golden ratio $\varphi^* = (\sqrt{5}-1)/2$, with value $g_{\max} = \frac{1}{2} \ln \varphi^* + \varphi^* \approx 0.3774$, and the boundary shifts to $\alpha_c(0) = 1 - g_{\max} \approx 0.6226$. Second, we describe three Monte Carlo schemes (direct, semismart, hybrid) and explain why a naive “smart” variant that refreshes its pattern cache fails near α_c . Third, we confirm the corrected boundary numerically at $N = 100$ by an escape-time measurement and by a ϕ residual heatmap.

2 Theory

Free energy split. For a chain in the basin of ξ^1 , the LSE sum is dominated by the $\mu = 1$ term, $H(\mathbf{x})/N \approx -\varphi_1$ with $\varphi_1 = \xi^1 \cdot \mathbf{x}/N$. Define the energy per spin relative to the retrieval baseline, $u(\varphi_1) = 1 - \varphi_1$, and the $(N-1)$ -sphere entropy per spin $s(\varphi_1) = \frac{1}{2} \ln(1 - \varphi_1^2)$. The retrieval free energy is the minimum of $f(\varphi_1, T) = u - Ts$:

$$f_{\text{ret}}(T) = (1 - \phi_{\text{eq}}) - \frac{T}{2} \ln(1 - \phi_{\text{eq}}^2), \quad \phi_{\text{eq}}(T) = \frac{1}{2}(-T + \sqrt{T^2 + 4}). \quad (3)$$

For a chain uncorrelated with ξ^1 ($\varphi_1 \approx 0$) the sum in (1) is dominated by the $M - 1$ spurious terms $S = \sum_{\mu \neq 1} \exp(N\varphi_{1\mu})$. Replacing the sum by an integral with density (2) and using $M = e^{\alpha N}$:

$$S \approx e^{\alpha N} \int_{-1}^1 \exp\left[N\left(\frac{1}{2} \ln(1 - \varphi^2) + \varphi\right)\right] d\varphi \equiv e^{\alpha N} \int e^{Ng(\varphi)} d\varphi. \quad (4)$$

$g(\varphi) = \frac{1}{2} \ln(1 - \varphi^2) + \varphi$ has an interior saddle at $g'(\varphi^*) = 0$, i.e. $\varphi^{*2} + \varphi^* - 1 = 0$, giving the golden ratio $\varphi^* = (\sqrt{5} - 1)/2 \approx 0.618$. Using the identity $1 - \varphi^{*2} = \varphi^*$ the saddle value is

$$g_{\max} = g(\varphi^*) = \frac{1}{2} \ln \varphi^* + \varphi^* \approx 0.3774. \quad (5)$$

Hence $N^{-1} \ln S \rightarrow \alpha + g_{\max}$ and the noise-floor energy per spin is $u_{\text{noise}}(\alpha) = 1 - (\alpha + g_{\max})$.

Boundary. Retrieval and noise floor cross at the boundary $\alpha_c(T)$:

$$\boxed{\alpha_c(T) = 1 - g_{\max} - f_{\text{ret}}(T), \quad \alpha_c(0) = 1 - g_{\max} \approx 0.6226.} \quad (6)$$

Compare with the Ramsauer Gaussian prediction $\alpha_c^G(0) = 1/2$. The two values differ by $g_{\max} - (1 - 1/\sqrt{2}) \approx 0.12$ per spin — a factor 1.25 in capacity.

Cusp picture. At fixed φ_1 the spurious sum evaluates to $e^{N(\alpha + g_{\max})}$ to leading order, independent of φ_1 . The leading-order free energy per spin reads

$$\frac{F(\varphi_1)}{N} = -\max(\varphi_1, \alpha + g_{\max}) - \frac{T}{2} \ln(1 - \varphi_1^2), \quad (7)$$

with a kink (“cusp”) at $\varphi_1 = \alpha + g_{\max}$. As T rises the basin minimum $\phi_{\text{eq}}(T)$ slides toward the cusp; at $\phi_{\text{eq}}(T) = \alpha + g_{\max}$ the local minimum disappears, defining the single-pattern spinodal

$$\alpha_c^{\text{sp}}(T) = \phi_{\text{eq}}(T) - g_{\max}, \quad T_{\text{sp}}(\alpha) = \frac{1 - (\alpha + g_{\max})^2}{\alpha + g_{\max}}. \quad (8)$$

Both α_c^{sd} and α_c^{sp} end at the same point $(1 - g_{\max}, 0) \approx (0.6226, 0)$. Figure 1 shows the free energy at $\alpha = 0.50$ for several T .

3 Monte Carlo schemes

All chains run on the $\sqrt{N}$ -sphere at $\beta_{\text{net}} = 1$ with Metropolis updates. The variants differ in how they handle the $M = e^{\alpha N}$ patterns.

Direct. Store all M patterns explicitly and evaluate (1) over all of them at every step. Memory and compute cost $\sim MN$. Feasible up to $N \approx 25$ at $\alpha \approx 0.7$ ($M \approx 4 \times 10^7$); impossible at $N = 100$, $\alpha = 0.6$ ($M \approx 10^{26}$).

Semismart. Keep only the patterns with overlap $\varphi_{1\mu} > \varphi_{\text{keep}}$ to the target ξ^1 . The cutoff is chosen so that the expected count above it equals a budget $K \sim 10^4$; the actual count is Poisson with parameter $M \cdot \Pr(\varphi > \varphi_{\text{keep}})$, and the K overlap values are drawn from a Beta tail. The discarded patterns enter the LSE log-sum as the analytic constant $\ln C_{\text{bulk}} = \alpha N + \ln \int_{-1}^{\varphi_{\text{keep}}} f(\varphi) e^{N\varphi} d\varphi$. The retained set is quenched — fixed once per disorder. We use semismart for the wide heatmap.

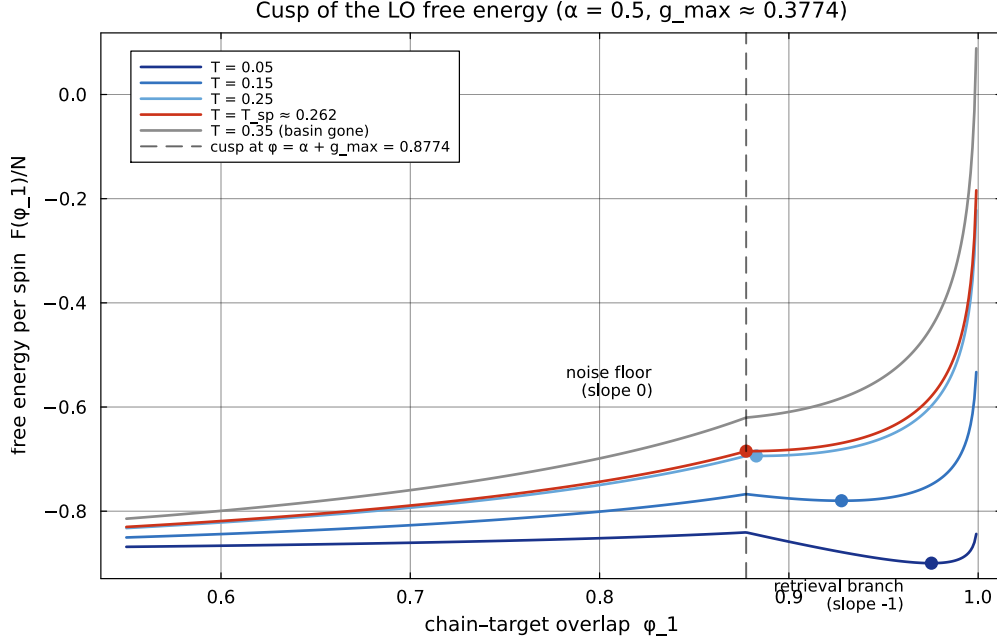


Figure 1: Leading-order free energy per spin $F(\varphi_1)/N$ from (7) at $\alpha = 0.50$ for several T . The cusp at $\varphi_1 = \alpha + g_{\max} \approx 0.877$ is the boundary between the retrieval branch (slope -1) and the noise-floor branch (slope 0). Dots: basin minima $\phi_{\text{eq}}(T)$. At $T = T_{\text{sp}}(0.50) \approx 0.262$ the minimum reaches the cusp; above this T the basin no longer exists.

Hybrid. A 1D Metropolis on the scalar φ_1 . Patterns with overlap $c_\mu > \varphi_c = \alpha + g_{\max}$ are sampled from the spherical Beta tail and kept fixed per disorder (count K_{cusp} , of order $M \cdot \Pr(\varphi > \varphi_c)$, in our $N = 100$ runs between ~ 4500 at $\alpha = 0.50$ and ~ 8500 at $\alpha = 0.62$). The contribution of the bulk ($c_\mu < \varphi_c$) is replaced by the φ_1 -independent constant $M e^{N g_{\max}}$ from (5). Hybrid is the right tool near α_c at $N = 100$, where direct is infeasible and the basin geometry is set by both the bulk floor and a small number of cusp-tail patterns.

Why naive “smart” fails. A faster variant — “smart” — refreshes its K kept patterns every N_{REFRESH} steps from a cone around the current chain. This breaks the quenched-disorder assumption. As the chain wanders, fresh patterns are minted at every refresh; the total set of distinct patterns seen along one trajectory exceeds M , and the chain effectively faces a larger pattern set than the model defines. Near the retrieval basin the extra competitors destroy the basin. Hybrid avoids the problem by fixing the cusp-tail set once per disorder and never refreshing it.

4 Results

We measure two observables at $N = 100$: the mean first-passage escape time τ from the retrieval basin, and the steady-state $\langle \varphi_1 \rangle - \phi_{\text{eq}}(T)$ residual.

Escape time. Figure 2 reports $\log_{10} \langle \tau \rangle$ vs T at $N = 100$ for five α values, averaged over 32 disorders, with a budget cap at 10^5 MC steps. Each curve is a basin escape profile: at low T the basin is stable for the full budget (upward triangles at the ceiling); above a critical $T_{\text{esc}}(\alpha)$ the escape time drops by several orders of magnitude over a T window of width ~ 0.01 .

T_{esc} moves to lower T as α rises. At $\alpha = 0.50$ the basin survives to $T \approx 0.085$; at $\alpha = 0.62$ it survives only to $T \approx 0.005$. Extrapolating to $\tau \rightarrow \infty$ at $T = 0$ gives a critical load at the right edge of the plot, near $\alpha = 1 - g_{\max} \approx 0.6226$.

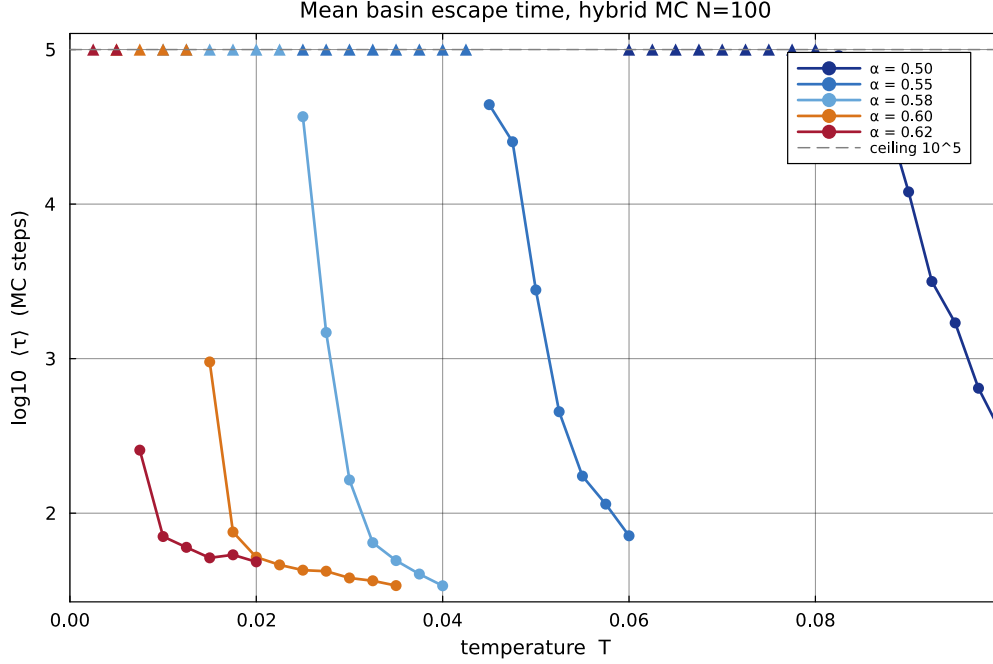


Figure 2: Mean basin escape time $\langle \tau \rangle$ vs T at $N = 100$ from hybrid MC, 32 disorders. Five α values: 0.50, 0.55, 0.58, 0.60, 0.62. Filled circles: mean over disorders that escaped within 10^5 MC steps. Upward triangles at $\log_{10} \tau = 5$: cells where no disorder escaped (lower bound only). The escape curve slides toward $T = 0$ as $\alpha \rightarrow 1 - g_{\max} \approx 0.6226$.

Residual heatmap. Figure 3 shows $\langle \varphi_1 \rangle - \phi_{\text{eq}}(T)$ over the (α, T) plane at $N = 100$ from the hybrid MC inside the grey rectangle, with honest MC at $N = 25$ and semismart MC at varying N outside. The basin (white) is bounded by a sharp transition (dark red) that follows the saddle line (6) and pinches at $\alpha \approx 0.6226$, $T = 0$.

5 Discussion

The Ramsauer threshold $\alpha_c = 1/2$ is a Gaussian artefact. The Gaussian density misses the integrand saddle at the golden ratio that the exact spherical density places at $\varphi^* = (\sqrt{5} - 1)/2$. Including it gives $\alpha_c = 1 - g_{\max} \approx 0.6226$, a factor 1.25 larger. The shift is per spin, not asymptotic: it appears at leading order in N and is visible already at $N = 100$.

Open items. (i) Hot init. All present runs start the chain at ξ^1 (cold). The basin boundary they trace is the spinodal, not the thermodynamic crossover; hot init at the same grid would delimit the metastability band. (ii) Cusp rounding at finite N . The leading-order $\max(\cdot, \cdot)$ in (7) is rounded on a scale $1/N$ in φ_1 . Near the endpoint $(0.6226, 0)$ the basin offset $\phi_{\text{eq}}(T) - (\alpha + g_{\max})$ falls inside the rounding window, and the leading-order spinodal slope needs an NLO correction. (iii) Heatmap consolidation. Figure 3 stitches three data sources. A clean version would use a single MC variant — semismart with a hybrid insert near the cusp — across the entire (α, T) plane.

References

- [1] Dmitry Krotov and John J. Hopfield. Dense associative memory for pattern recognition. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 29, pages 1172–1180, 2016.
- [2] Hubert Ramsauer, Bernhard Schäffl, Johannes Lehner, Philipp Seidl, Michael Widrich, Lukas

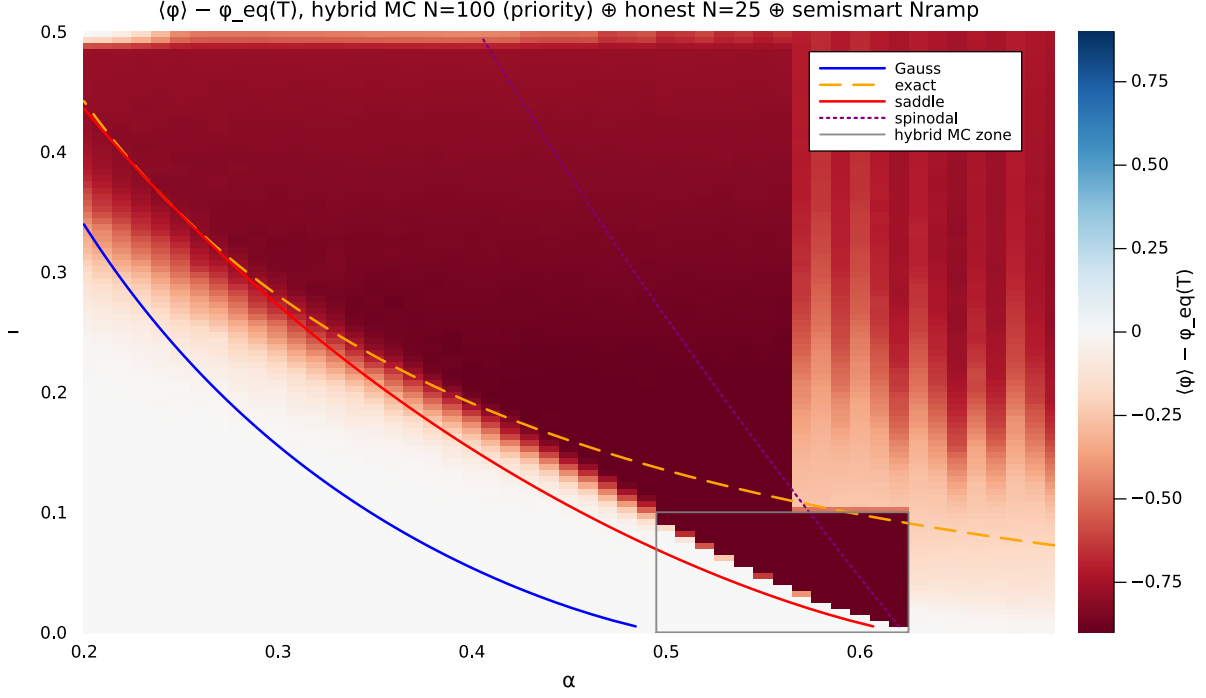


Figure 3: $\langle \varphi_1 \rangle - \phi_{\text{eq}}(T)$ from Monte Carlo over the (α, T) plane. Curves: α_c^G (blue, Ramsauer Gaussian), α_c^{sd} (red, this work, exact density saddle), α_c^{sp} (purple, spinodal). Inside the grey rectangle: hybrid MC at $N = 100$ on a fine T grid; outside: honest MC at $N = 25$ ($\alpha \leq 0.56$) and semismart MC at varying N ($\alpha \leq 0.70$), linearly interpolated to the same fine T grid. The basin pinches at $\alpha \approx 0.6226$, not 0.5. A later revision will replace this composite with a single honest + semismart + hybrid-insert build.

Gruber, Markus Holzleitner, Thomas Adler, David Kreil, Michael K. Kopp, Günter Klambauer, Johannes Brandstetter, and Sepp Hochreiter. Hopfield networks is all you need. In *International Conference on Learning Representations*, 2021.